

0x03BF (omi)

Omi-Matrix: The Anatomy of a System

From Mathematical Blueprint
to Physical Open-World Portal

0x039F (imo)

0x03BF (omi)

0x039F (imo)

$$Q_{xy} = 60x^2 + 16xy + 4y^2$$

$4y^2$ | The Low Gate (OMI)

Localized atomic data,
feature bits, CSS
attribute selectors.

$16xy$ | The Cross Bridge

The active execution
matrix where code meets
DOM, traversal order.

$60x^2$ | The High Gate (IMO)

Sexagesimal spatial
orientation, CSSOM
access, Identity.

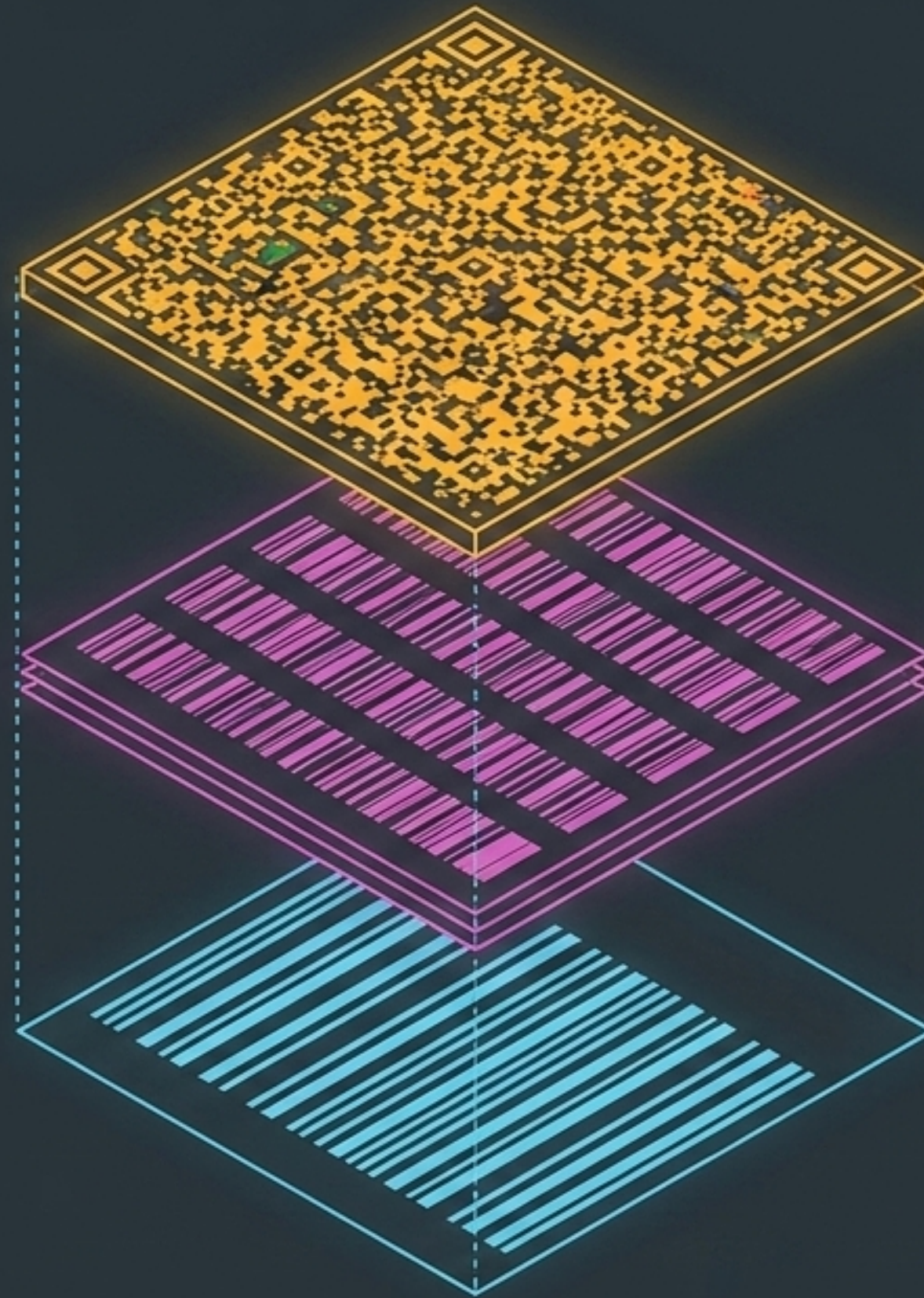
0x03BF (omi)

0x039F (imo)

0x03BF (omi)

0x039F (imo)

Middle Layer |
Code16K Stacked
Represents 16xy
Bridge-plane, linear
replay bitboard, omi
-> payload -> imo.



Top Layer |
JABCode Color Matrix
Represents $60x^2$
High-plane identity,
RGB/synset projection.

Bottom Layer |
Code128 Linear
Represents $4y^2$
Low-plane atomic
transport lane, data-*
values.

0x03BF (omi)

0x039F (imo)

0x03BF (omi)

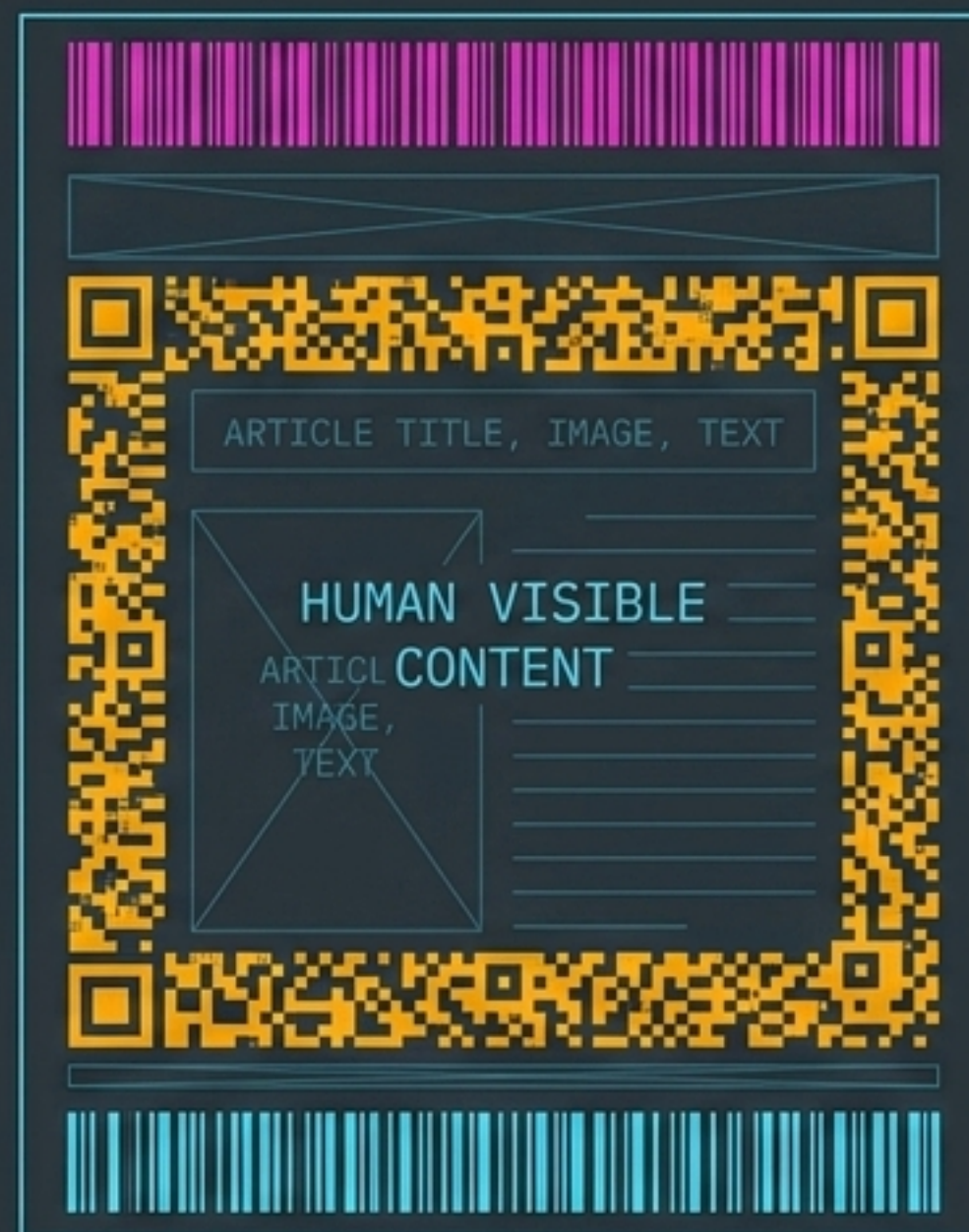
0x039F (imo)

Math Term	Plane Name	Barcode Symbology	Payload Role	OMI Meaning
$4y^2$	Low Plane	Code128	Atomic Lane	feature bits, POS flags, data-*
$16xy$	Bridge Plane	Code16K	Replay Bitboard	row transport, o---o joining, linear traversal
$60x^2$	High Plane	JABCode	Color Matrix	object identity, SpectrumDOM, bounding projection

0x03BF (omi)

0x039F (imo)

The Optical Page Frame



← Header: Code16K acting as the DOM Header and replay strip.

← Body: Human-visible content wrapped by the JABCode color matrix acting as the DOM Body.

← Footer: Code128 acting as the low-edge rail and boundary footer.

The page frame allows a document to be read in two ways: Human Reading (visible document) and Machine Scanning (OMI carrier envelope).

The Encapsulation Rule

omi (0x03BF) - Greek small letter omicron. The Low Gate / Entry point.

imo (0x039F) - Greek capital letter omicron. The High Gate / Exit point.



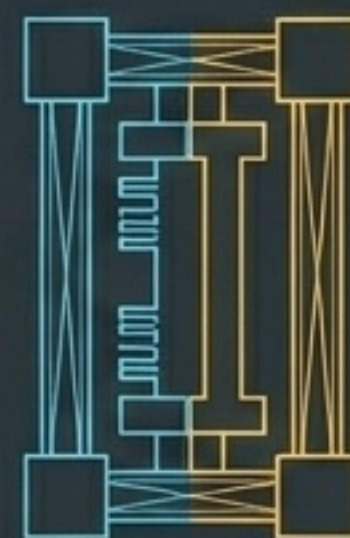
00000000	00000000	00000000	00000000	00000000	00000000
00000000	00000000	00000000	00000000	00000000	00000000
00000000	00000000	00000000	00000000	00000000	00000000
00000000	00000000	00000000	00000000	00000000	00000000
00000000	00000000	00000000	00000000	00000000	00000000
00000000	00000000	00000000	00000000	00000000	00000000

— — —

00000000	00000000	00000000	00000000	00000000	00000000
00000000	00000000	00000000	00000000	00000000	00000000
00000000	00000000	00000000	00000000	00000000	00000000
00000000	00000000	00000000	00000000	00000000	00000000
00000000	00000000	00000000	00000000	00000000	00000000
00000000	00000000	00000000	00000000	00000000	00000000

— — —

00000000	00000000	00000000	00000000	00000000	00000000
00000000	00000000	00000000	00000000	00000000	00000000
00000000	00000000	00000000	00000000	00000000	00000000
00000000	00000000	00000000	00000000	00000000	00000000
00000000	00000000	00000000	00000000	00000000	00000000
00000000	00000000	00000000	00000000	00000000	00000000



Payload Bus

120 characters of hyphen-delimited 4-character hex blocks. Mappable directly to an IPv6 source address.

Concept: Everything is exactly 128 characters long. Nothing enters or exits without these escape sequences.

The Base-Spec Router Map

0x039F (imo)

0x03BF (omi)



0x03BF (omi)

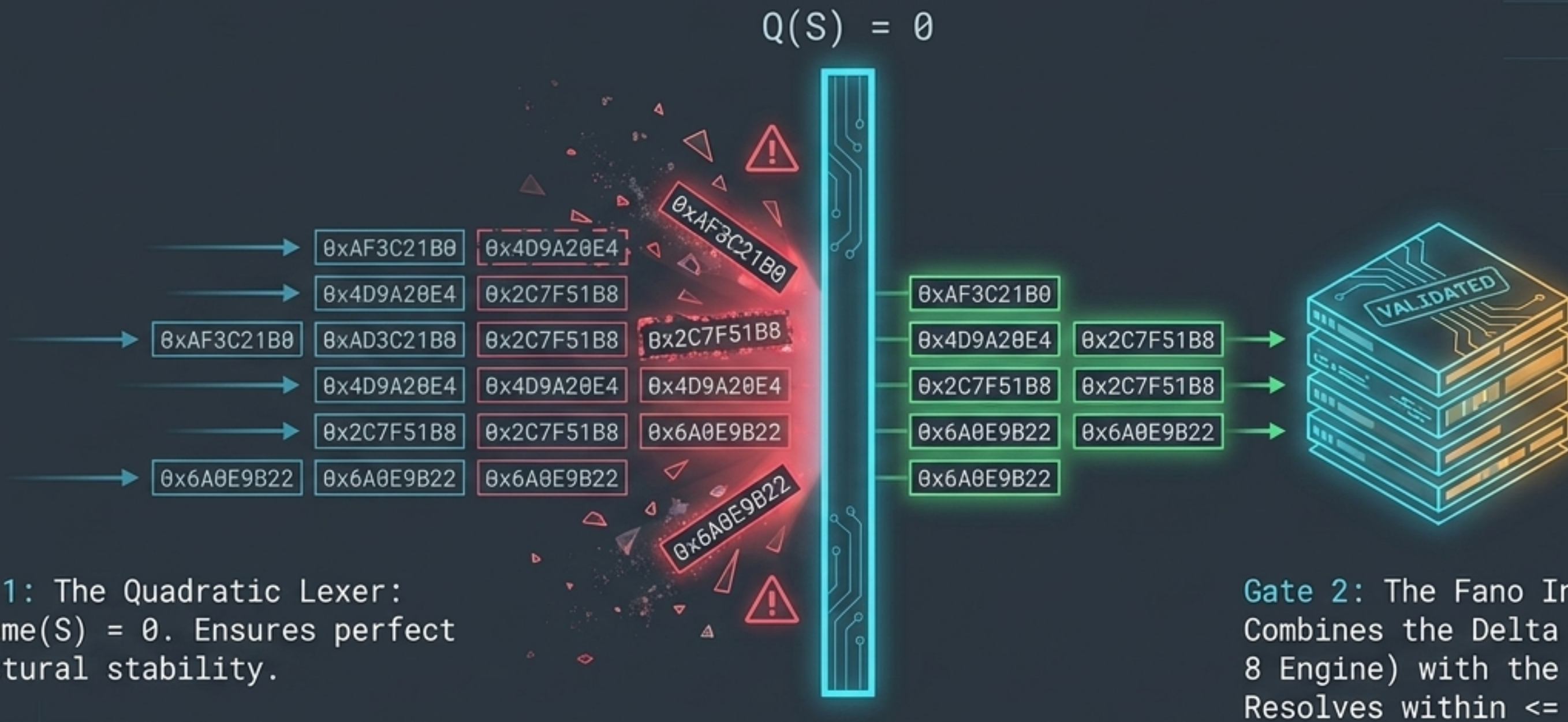
A memory-mapped bitboard carrier route. No central database required.
All identity comes from the validated frame, not the carrier.

0x039F (imo)

The Object Agreement Pipeline



The Validation Gate



Takeaway: Invalid frames are dropped at the NIC driver level before they ever touch userspace.

The Ring Indexer & Receipt Ledger



Concept: OMI's memory is a fixed circular buffer of 5040 slots (exactly 7!).

Slot Anatomy:

Each 64-bit slot stores:

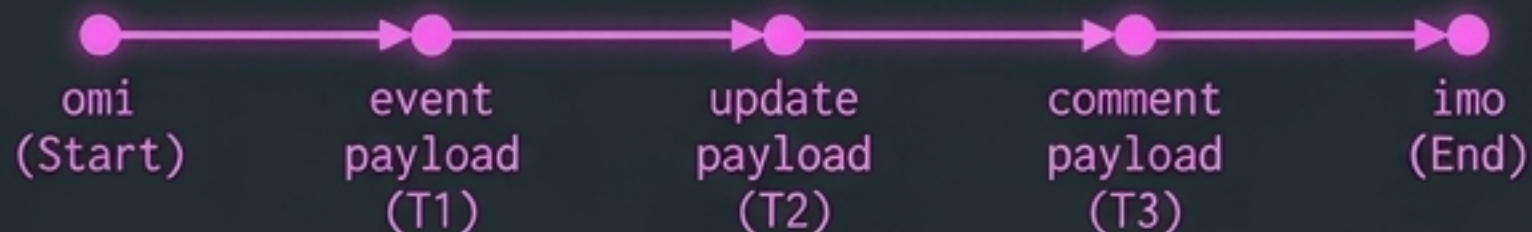
- Provenance (16 bits)
- Steps (8 bits)
- LL (8 bits)
- NN (16 bits)
- MM (16 bits)

Mechanic:

Memory advancement is a function of orbital distance, not wall-clock seconds.

The Dual Inspection Views

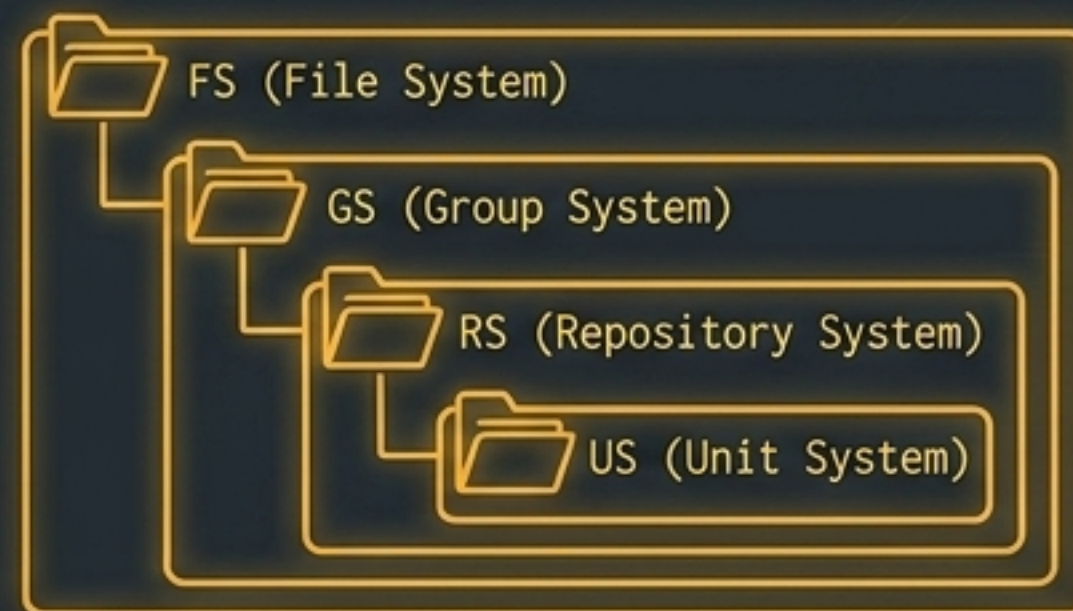
Linear Replay View



What happened?
What came before this?
What does this replay from?
What is the source/object flow?

omi -> event payload -> update payload
-> comment payload -> imo

Hierarchical Containment View



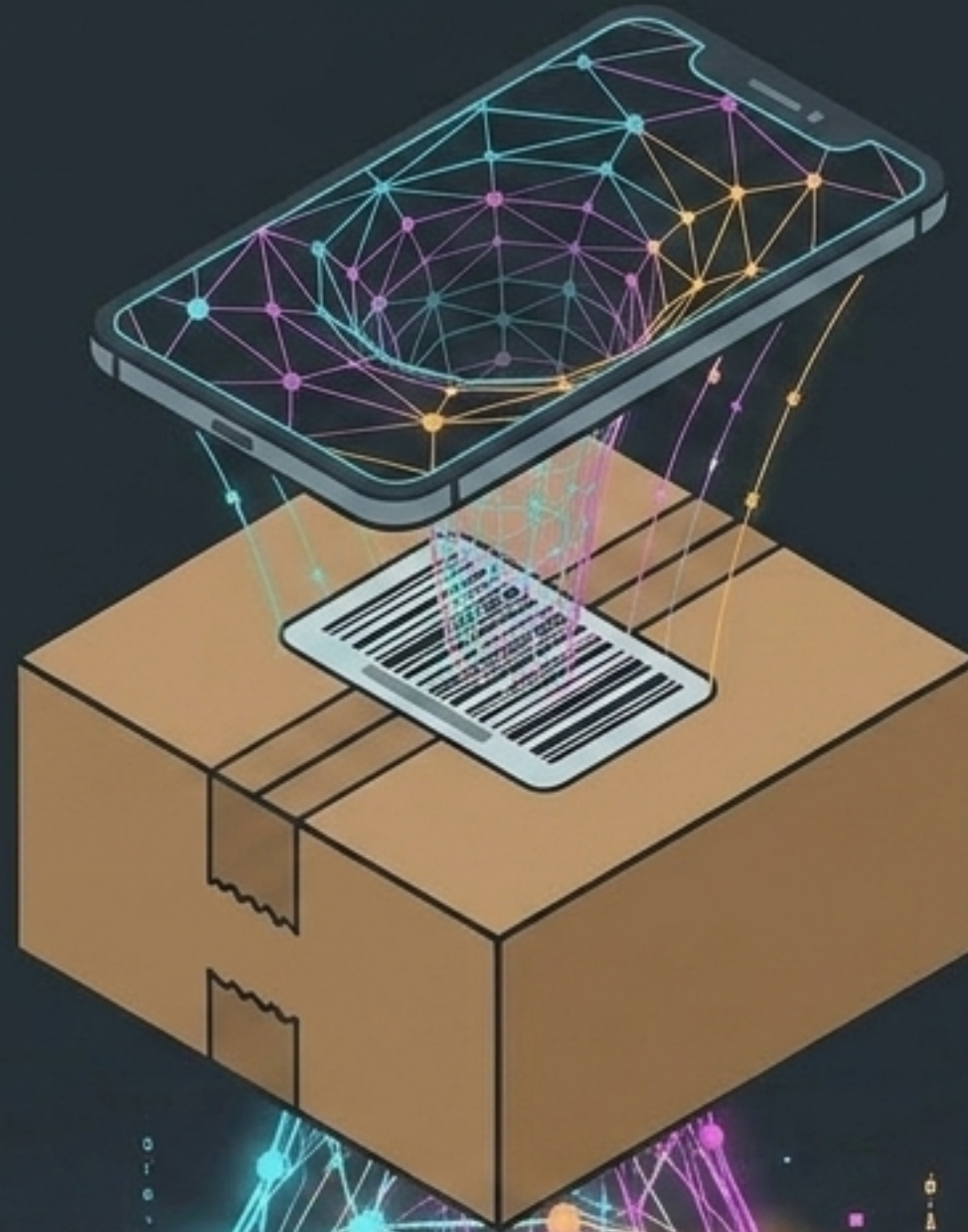
Where does this belong?
What contains it?
What context controls it?
What unit is being inspected?

FS -> GS -> RS -> US

The protocol preserves the proof path for both timeline traversal and structural containment.

The Zero-Constant Projection

$$P_{\text{matrix}}(0,0) = 0$$

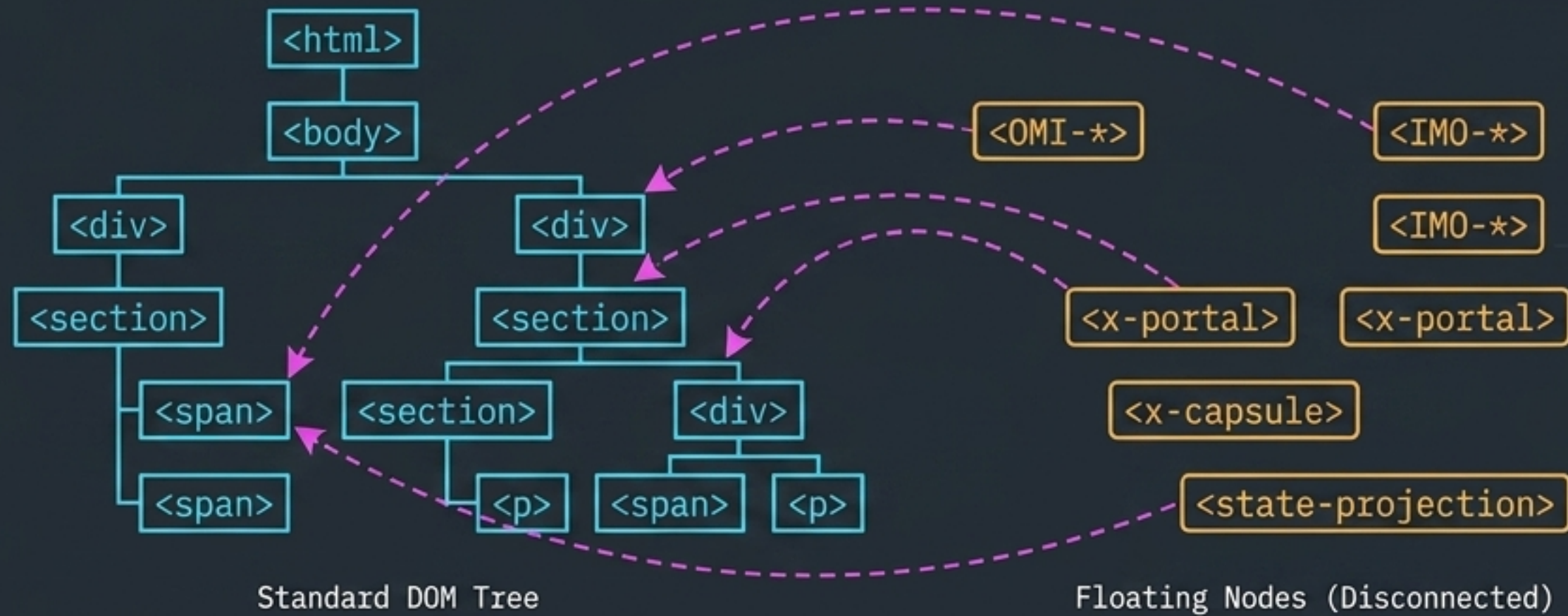


A barcode cannot smuggle truth.

No hidden identity can live in an additive constant.

The physical Omi-Matrix turns any surface into a literal, trustless portal to a decentralized object graph.

Wormhole Portals & State Teleportation



Concept

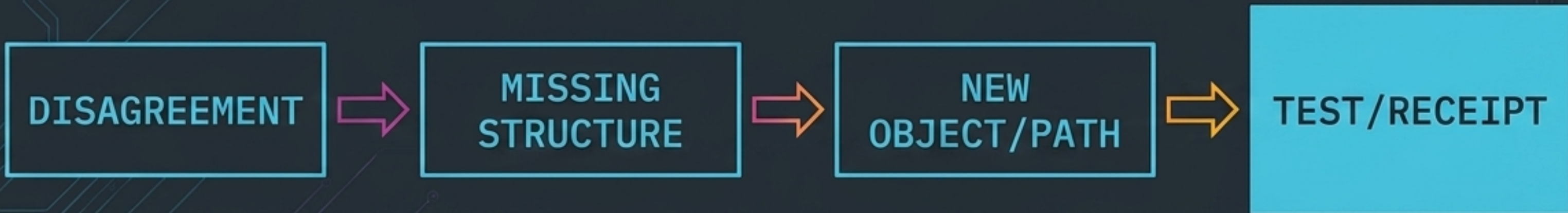
Standard trees enforce parent-child relationships. OMI frames need to teleport state across disconnected contexts.

Execution

Floating <OMI-*> and <IMO-*> nodes act as projection capsules. They bypass the standard tree via ShadowDOM, SVG <use>, and encapsulated iframes.

Disagreement as an Unresolved Path

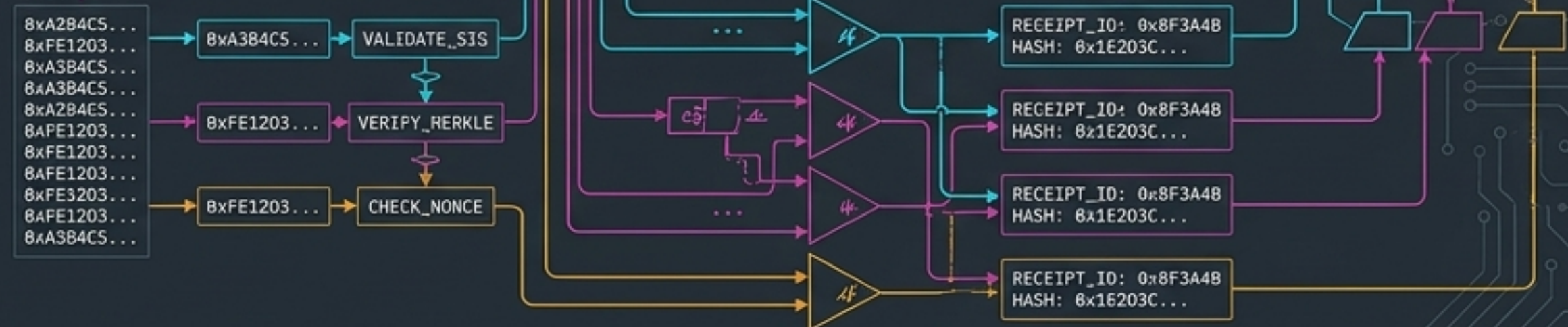
In a closed platform, disagreement becomes moderation or deletion. In OMI, disagreement is treated as missing structure.



HUMAN PERCEPTION

MACHINE REALITY

**SIMPLE LABELS FOR HUMANS.
RECEIPTS FOR MACHINES.
REPLAY FOR VERIFICATION.**



The consumer experiences an open-world web portal; the protocol sees validated state.

Identity is Object-Based, Not Platform-Owned

Event Card

TITLE

Global Climate Summit

SUMMARY

Annual conference on environmental sustainability.

AUTHOR/SOURCE

United Nations / @UN_Environment

TIME

2024-05-15 09:00 UTC

LOCATION/CONTEXT

New York City, NY, USA / Convention Center

STATUS

Confirmed

ACTIONS

View Details, Share, Add to Calendar

omi address: 0x03BF34A8C5...

source path	0x03BF34A8C5...
-------------	-----------------

object path	/USA/NY/NYC/Convention_Center/Event_123
-------------	---

object path	/USA/NY/NYC/Convention_Center/Event_123
-------------	---

	2024-05-15 09:00 UTC
--	----------------------

hierarchical containment:	/USA/NY/NYC/Convention_Center/Event_123
---------------------------	---

object path	0x03BF34A8C5...
-------------	-----------------

frame receipt:	0x1E2D3C... (VALIDATED)
	(VALIDATED)

linear replay:	[VIEW, SHARE, ADD]
----------------	----------------------

0x03BF (omi)

0x039F (imo)

An open-world portal
does not own truth. It
projects accepted objects.

The OMI open-world portal lets people browse
ordinary web objects while preserving extraordinary
cryptographic structure underneath.

0x03BF (omi)

0x039F (imo)